

Perhaps unfortunately, AFS is likely on the decline. Because NFS became an open standard, many different vendors supported it, and, along with CIFS (the Windows-based distributed file system protocol), NFS dominates the marketplace. Although one still sees AFS installations from time to time (such as in various educational institutions, including Wisconsin), the only lasting influence will likely be from the ideas of AFS rather than the actual system itself. Indeed, NFSv4 now adds server state (e.g., an “open” protocol message), and thus bears an increasing similarity to the basic AFS protocol.

## References

[B+91] “Measurements of a Distributed File System”

Mary Baker, John Hartman, Martin Kupfer, Ken Shirriff, John Ousterhout  
SOSP '91, Pacific Grove, California, October 1991

*An early paper measuring how people use distributed file systems. Matches much of the intuition found in AFS.*

[H+11] “A File is Not a File: Understanding the I/O Behavior of Apple Desktop Applications”

Tyler Harter, Chris Dragga, Michael Vaughn,  
Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau  
SOSP '11, New York, New York, October 2011

*Our own paper studying the behavior of Apple Desktop workloads; turns out they are a bit different than many of the server-based workloads the systems research community usually focuses upon. Also a good recent reference which points to a lot of related work.*

[H+88] “Scale and Performance in a Distributed File System”

John H. Howard, Michael L. Kazar, Sherri G. Menees, David A. Nichols, M. Satyanarayanan,  
Robert N. Sidebotham, Michael J. West  
ACM Transactions on Computing Systems (ACM TOCS), page 51-81, Volume 6, Number 1,  
February 1988

*The long journal version of the famous AFS system, still in use in a number of places throughout the world, and also probably the earliest clear thinking on how to build distributed file systems. A wonderful combination of the science of measurement and principled engineering.*

[R+00] “A Comparison of File System Workloads”

Drew Roselli, Jacob R. Lorch, Thomas E. Anderson  
USENIX '00, San Diego, California, June 2000

*A more recent set of traces as compared to the Baker paper [B+91], with some interesting twists.*

[S+85] “The ITC Distributed File System: Principles and Design”

M. Satyanarayanan, J.H. Howard, D.A. Nichols, R.N. Sidebotham, A. Spector, M.J. West  
SOSP '85, Orcas Island, Washington, December 1985

*The older paper about a distributed file system. Much of the basic design of AFS is in place in this older system, but not the improvements for scale.*

[V99] “File system usage in Windows NT 4.0”

Werner Vogels

SOSP '99, Kiawah Island Resort, South Carolina, December 1999

*A cool study of Windows workloads, which are inherently different than many of the UNIX-based studies that had previously been done.*